Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES – (24SDCS03)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	EduFusion AI: An Intelligent Agentic and Microservices-Based Framework for Smart Campus Optimization	
S. No.	**University ID**	**Name**
1	2420030381	B V S JAYADEEP
2	2420030588	CHINTALAPATI BHARGAVI DEVI
3	2420030499	SAKHAMURI ANJANA
Name of the Guide		**Dr. Srikanth Cherukuvada**

Abstract:

Modern educational institutions generate and process large volumes of heterogeneous data related to student activities, academic operations, infrastructure utilization, transportation, events, communication, and administrative services. Conventional campus management systems generally operate as isolated applications, resulting in fragmented information, delayed decision-making, limited automation, and inefficient utilization of institutional resources. To address these challenges, this project proposes **EduFusion AI**, an intelligent agentic and microservices-based framework designed to optimize and automate smart campus operations through coordinated Artificial Intelligence agents and scalable cloud-native services.

The proposed framework integrates multiple specialized AI agents responsible for different campus functions, including student assistance, academic support, resource allocation, event coordination, infrastructure monitoring, notification management, and administrative decision support. Instead of relying on a single centralized intelligence layer, EduFusion AI employs an **agentic architecture** in which autonomous agents can analyze contextual information,

communicate with relevant services, determine appropriate actions, and coordinate with other agents to accomplish complex campus-level tasks. This enables the system to move beyond simple query-based assistance toward intelligent, context-aware, and goal-oriented automation.

The application is implemented using a **microservices architecture**, where individual campus functionalities are developed as independently deployable and scalable services. This approach improves modularity, fault isolation, maintainability, and scalability while allowing different services to communicate through well-defined APIs and event-driven mechanisms. Real-time communication and asynchronous processing can be incorporated to enable rapid responses to changing campus conditions, such as classroom availability, resource utilization, student requests, event updates, and infrastructure alerts. An API gateway and centralized service management layer provide controlled access, service coordination, authentication, and monitoring across the distributed system.

EduFusion AI further incorporates data processing and intelligent decision-making mechanisms to analyze campus information and generate actionable insights. By combining agentic reasoning with real-time service interaction, the framework can identify operational requirements, prioritize tasks, recommend suitable actions, and automate repetitive workflows. The system is designed to support students, faculty, and administrators through a unified intelligent platform while reducing manual intervention and improving operational efficiency.

The proposed framework demonstrates how **Artificial Intelligence, intelligent agents, microservices, real-time communication, and cloud-native technologies** can be integrated to create a scalable smart-campus ecosystem. The expected outcome is a flexible and intelligent platform capable of improving resource utilization, service responsiveness, decision-making, and overall campus experience. EduFusion AI provides a foundation for transforming conventional campus management into a more connected, autonomous, data-driven, and intelligent smart-campus environment.

Project Architecture/ Project Overview:

